

# The Hunger Games Analysis



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# Table Of Contents

01 Introduction and Research Overview

02 Book 1 *The Hunger Games* Analysis

03 Book 2 *Catching Fire* Analysis

04 Book 3 *Mockingjay* Analysis

05 Cross Book Comparisons

06 Comparison to LLM

07 Statistical Analysis

08 Interpretations & Conclusions



01

# Introduction

# Introduction

- We are analyzing *The Hunger Games* trilogy by Suzanne Collins:
  - *The Hunger Games*
  - *Catching Fire*
  - *Mockingjay*
- This trilogy has had a large global impact on dystopian fiction media & communicates themes still relevant today
- We are interested in the character, thematic, and linguistic progression within each book and throughout the trilogy for this project

# Data & Study Design

## Data:

- Downloaded the 3 novels in pdf format from the internet
- Split the novels into chapters (27 per book, 81 documents total)
- Preprocessed the text

## Key Measures:

- Word Frequency
- Sentiment
- TF-IDF
- Topic Modeling

# Research Questions

- 1. How does sentiment shift across the three novels, and does it become more negative as the narrative progresses?**
- 2. How do word frequency patterns change from novel to novel, particularly regarding terms related to survival, violence, power, and rebellion?**
- 3. Does the usage of certain words increase as the narrative evolves from survival to political revolution?**
- 4. How does Suzanne Collins writing differ across each book?**

02

Book 1 *The  
Hunger Games*  
Analysis

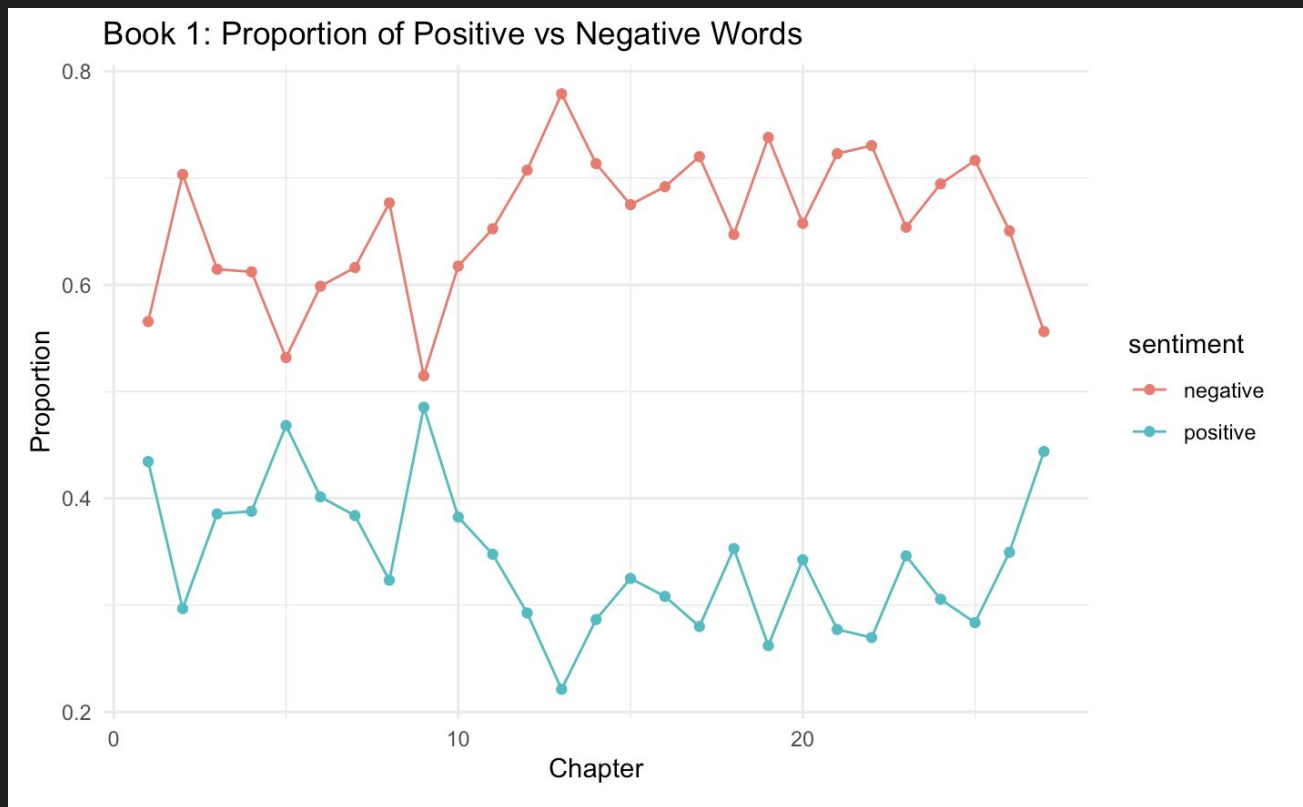
THE  
HUNGER  
GAMES



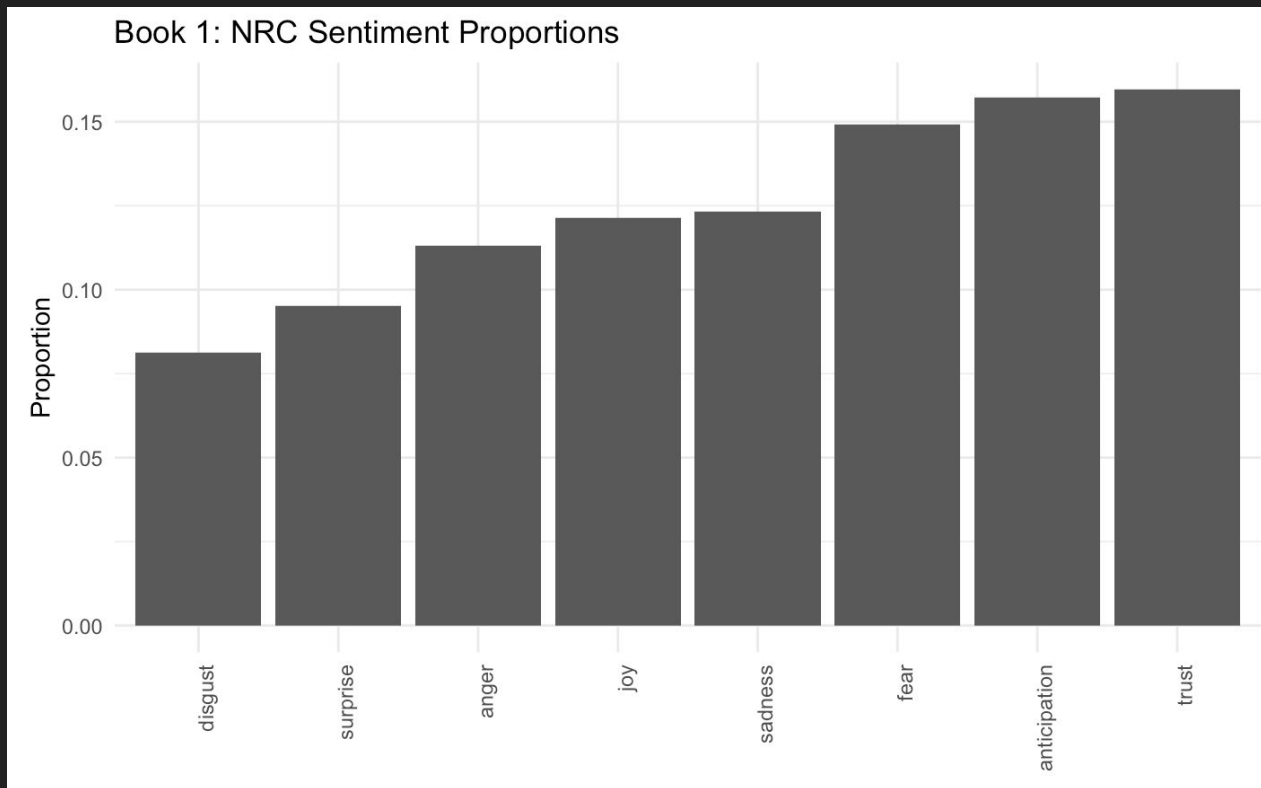
*New York Times Bestselling Author*

SUZANNE  
COLLINS

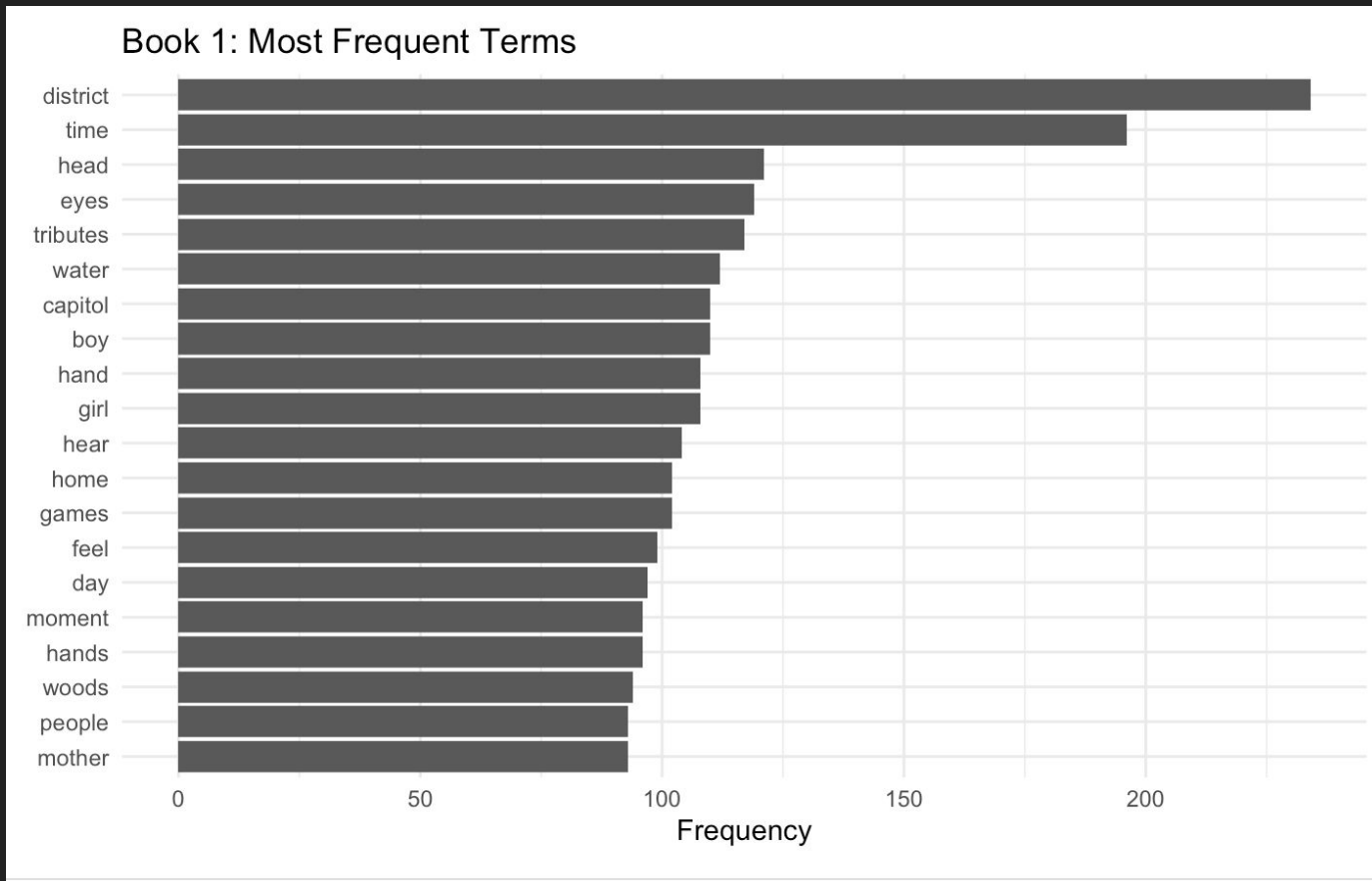
# Book 1 Sentiment Over Time



# Book 1 NRC Sentiments

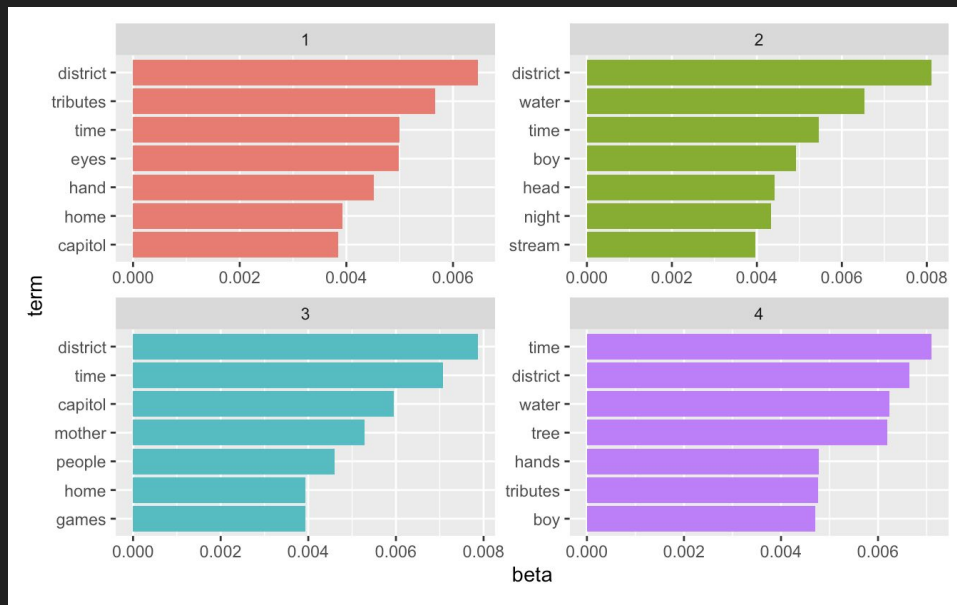


# Book 1 Vocabulary



# Book 1 Themes Detected via Topic Modeling

- Topic 1: The Games & Physical Experience
- Topic 2: Survival in the Arena
- Topic 3: Home & Society
- Topic 4: The Arena & Combat



The Second Book of THE HUNGER GAMES

# CATCHING FIRE



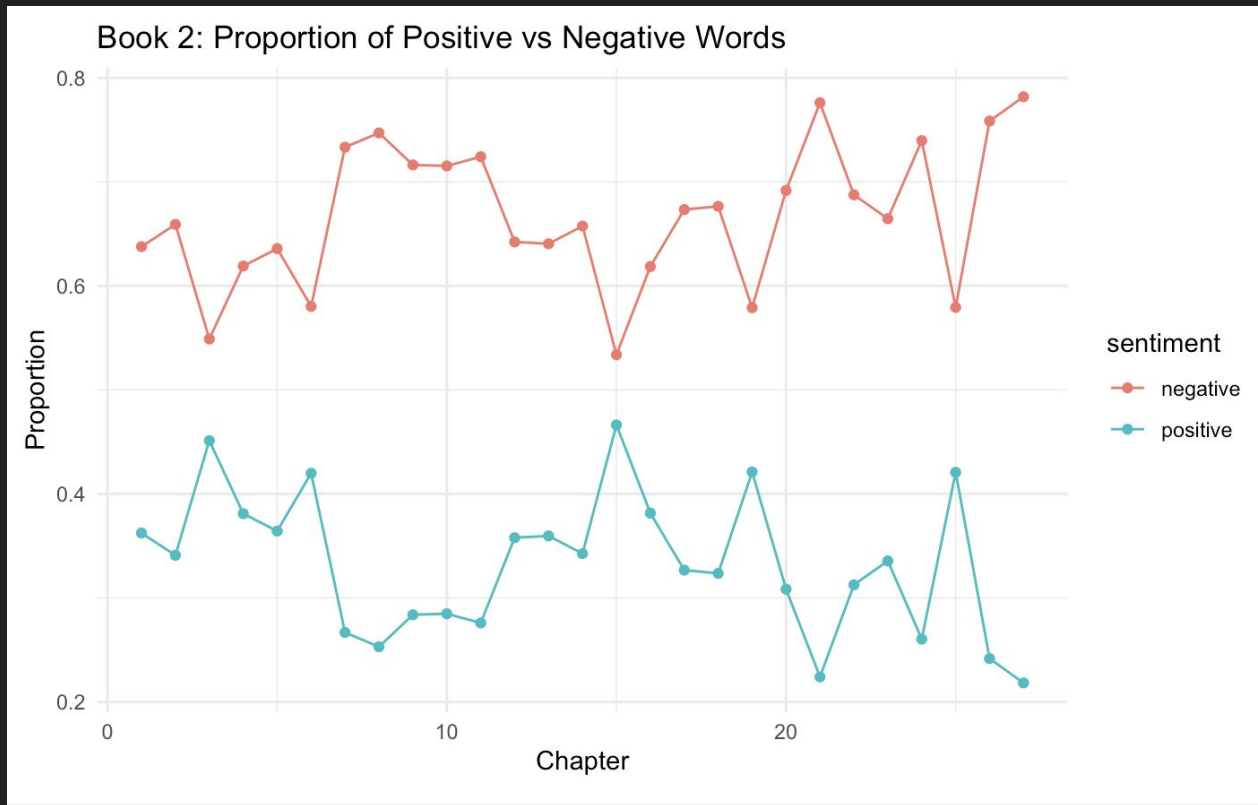
New York Times Bestselling Author

SUZANNE  
COLLINS

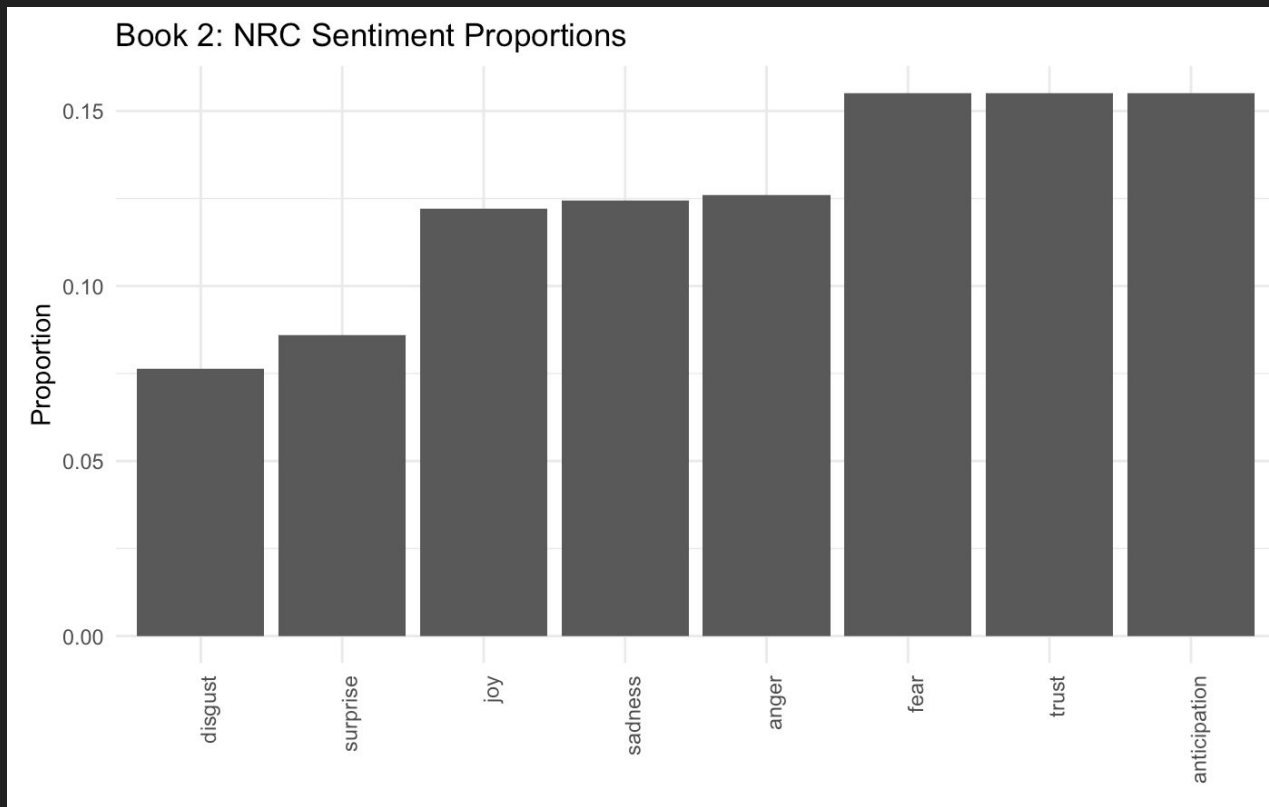
## 03

Book 2 *Catching  
Fire* Analysis

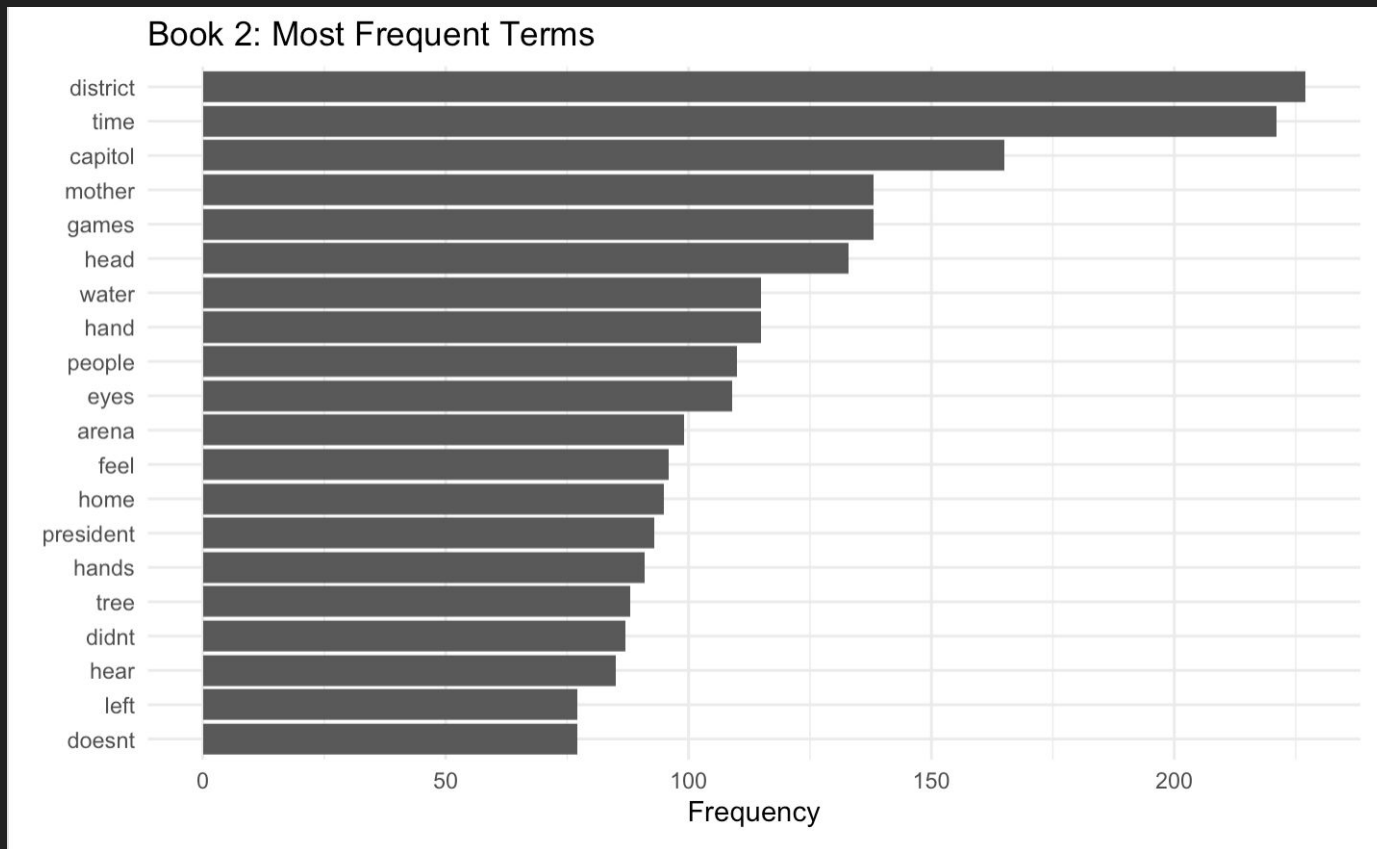
# Book 2 Sentiment Over Time



# Book 2 NRC Sentiments

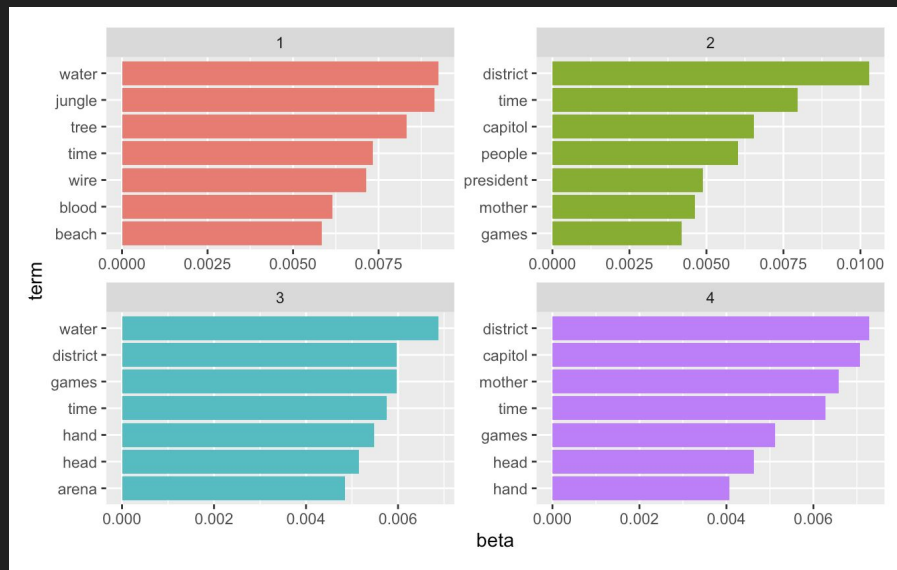


# Book 2 Vocabulary



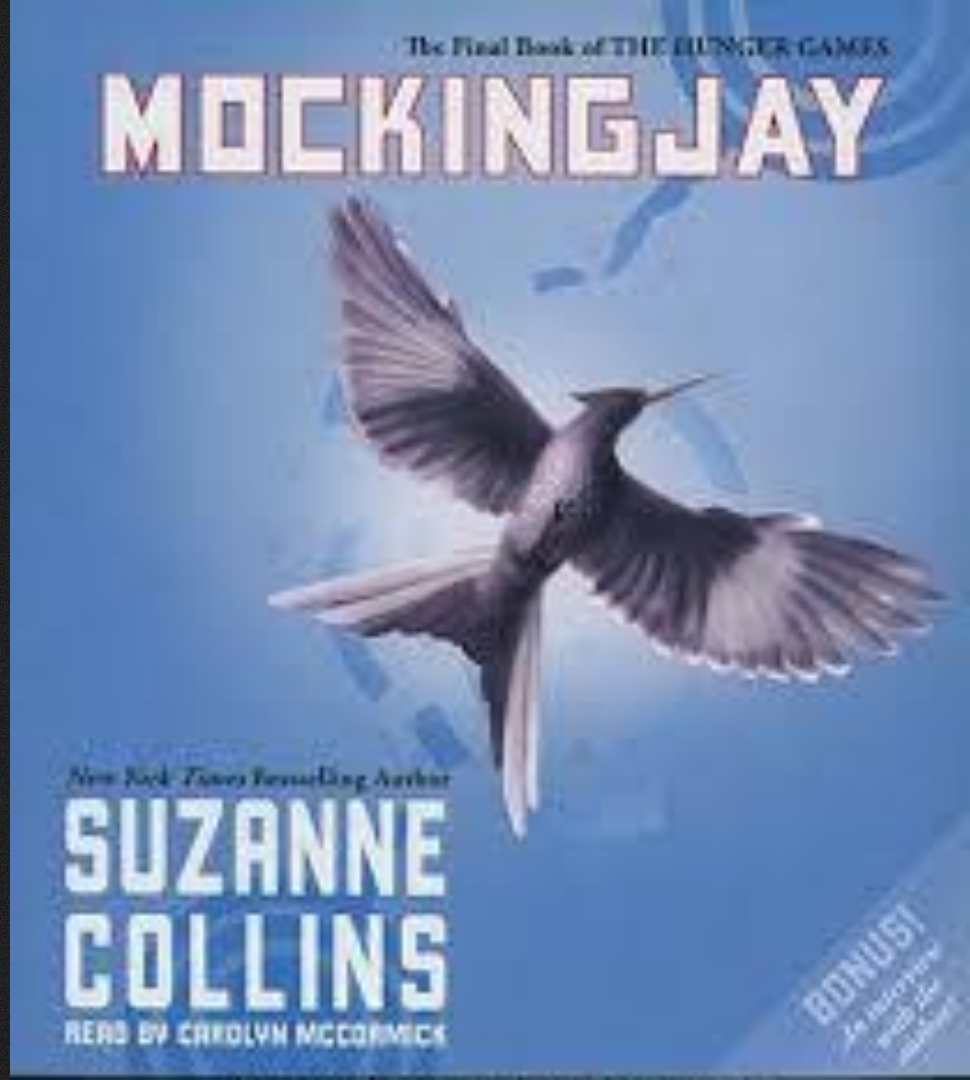
# Book 2 Themes Detected via Topic Modeling

- Topic 1: The Quarter Quell Arena
- Topic 2: Political Unrest
- Topic 3: Arena Combat
- Topic 4: Personal & Political Ties

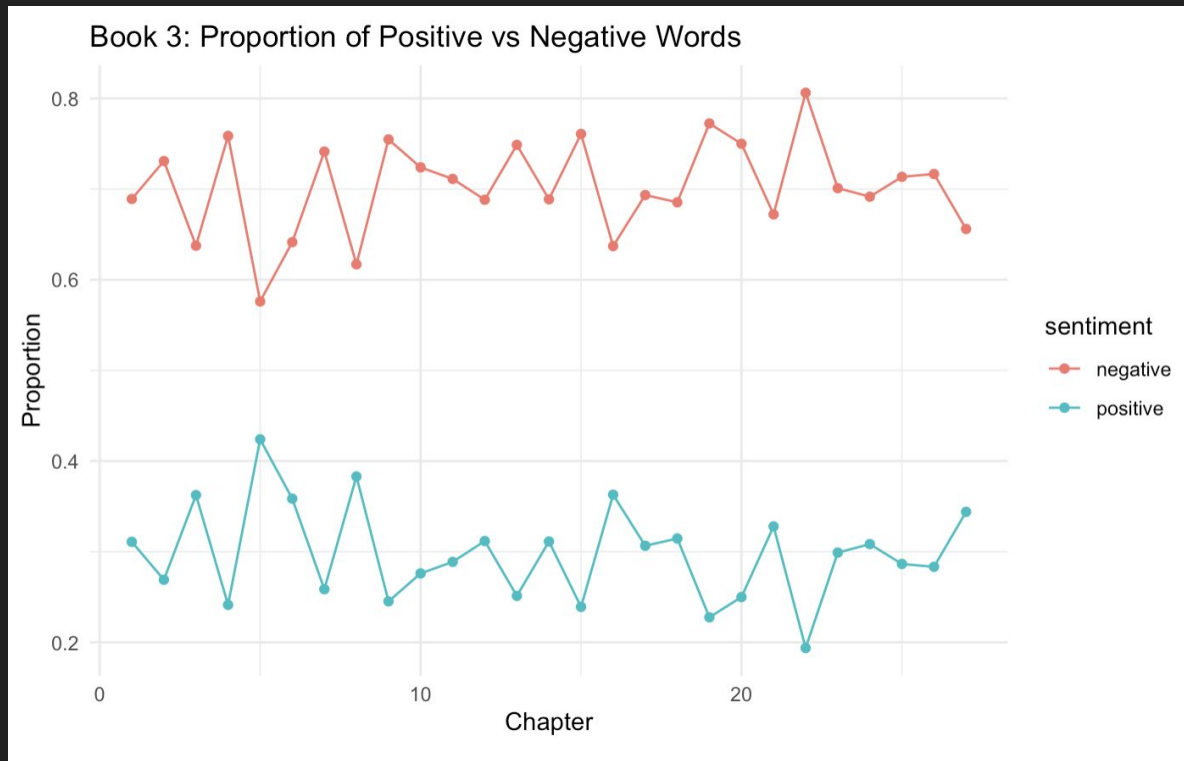


04

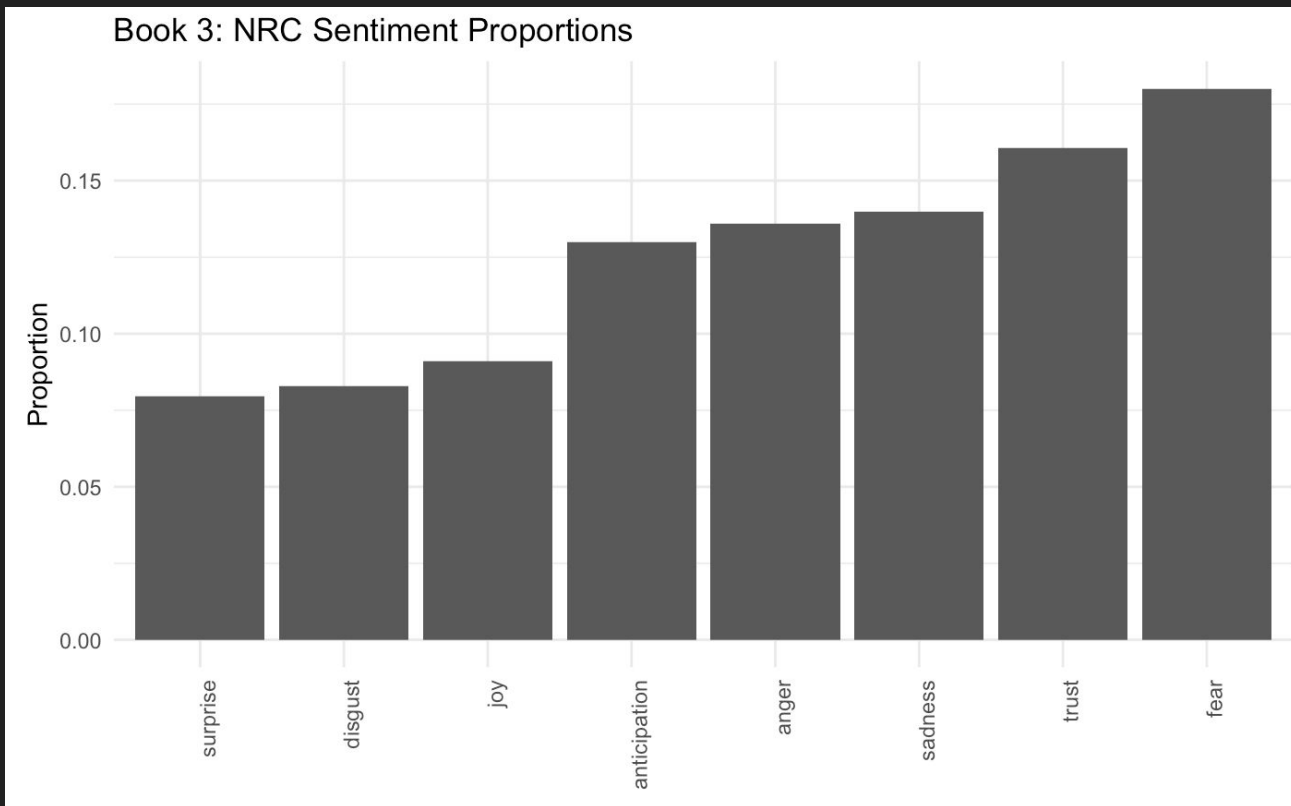
Book 3  
*Mockingjay*  
Analysis



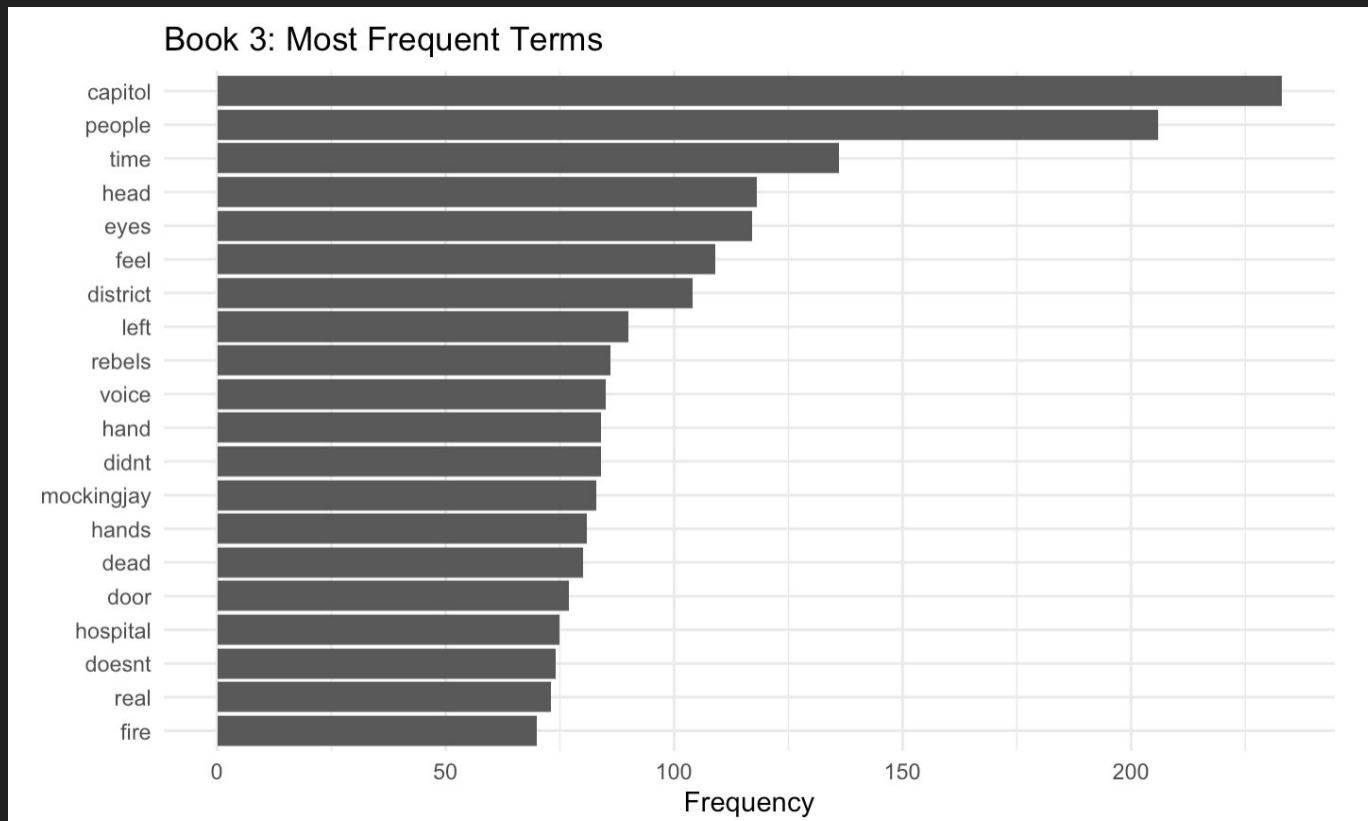
# Book 3 Sentiment Over Time



# Book 3 NRC Sentiments

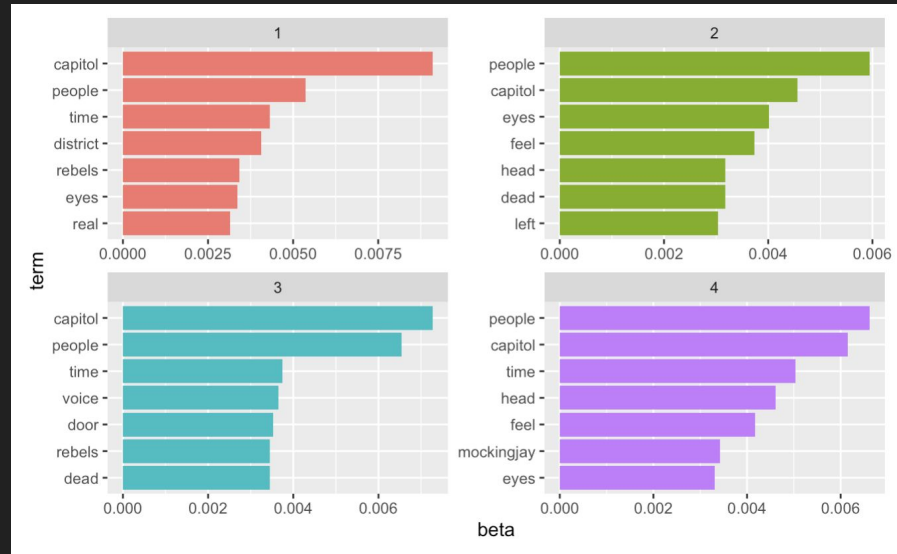


# Book 3 Vocabulary



# Book 3 Themes Detected via Topic Modeling

- Topic 1: The Rebellion & War Effort
- Topic 2: Psychological Trauma
- Topic 3: Assault on the Capital
- Topic 4: Symbolism of the Mockingjay





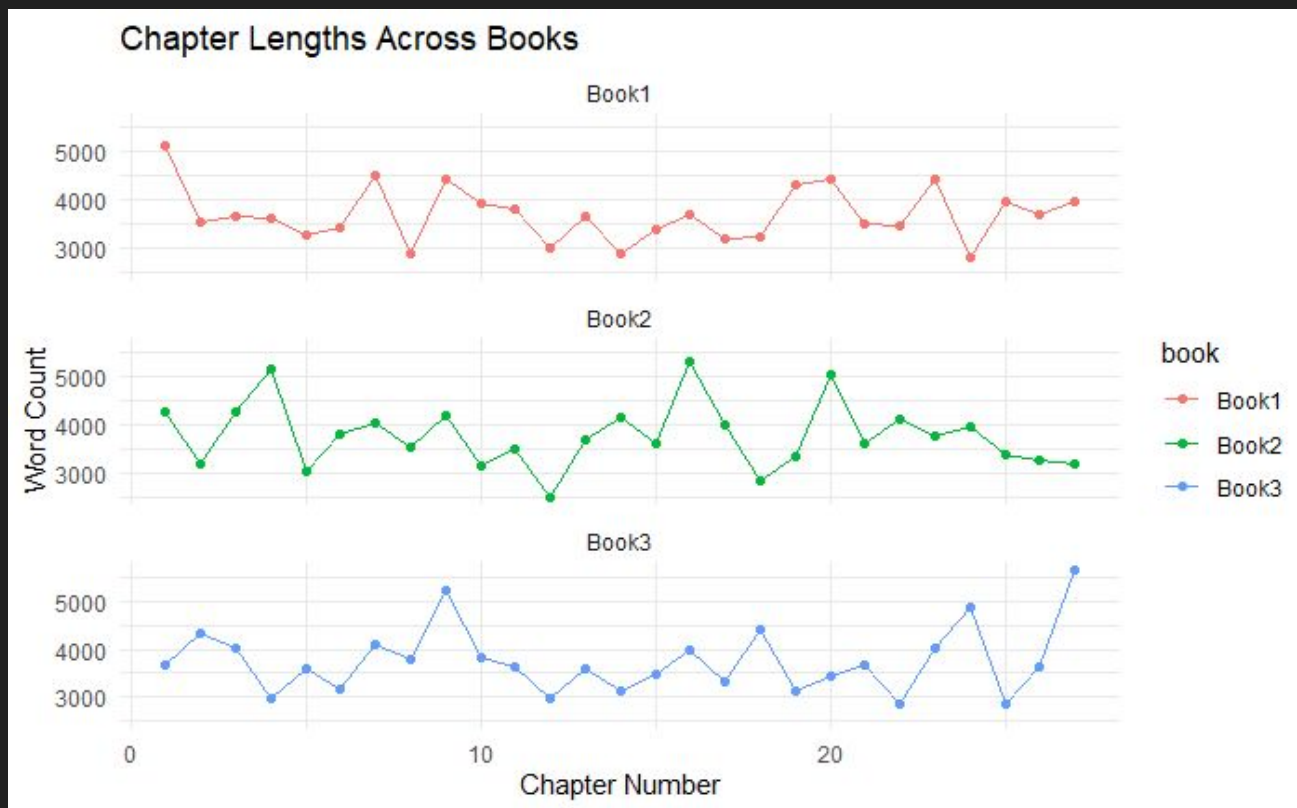
05

## Cross Book Comparisons

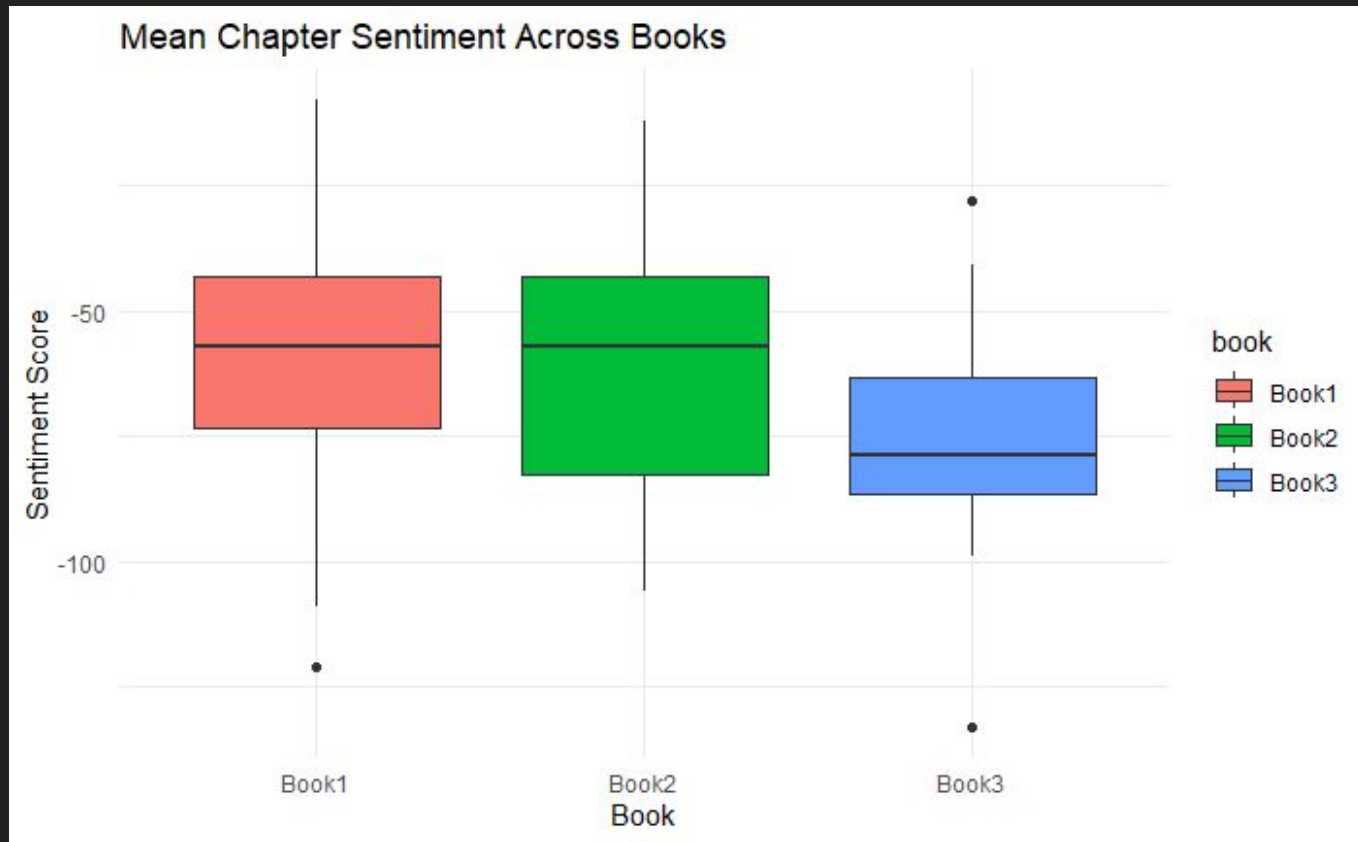
## Summary Statistics

| Book          | Total Words | Unique Words | Lexical Diversity |
|---------------|-------------|--------------|-------------------|
| Hunger Games  | 99893       | 7544         | 7.55%             |
| Catching Fire | 101719      | 7807         | 7.68%             |
| Mockingjay    | 101458      | 8634         | 8.51%             |

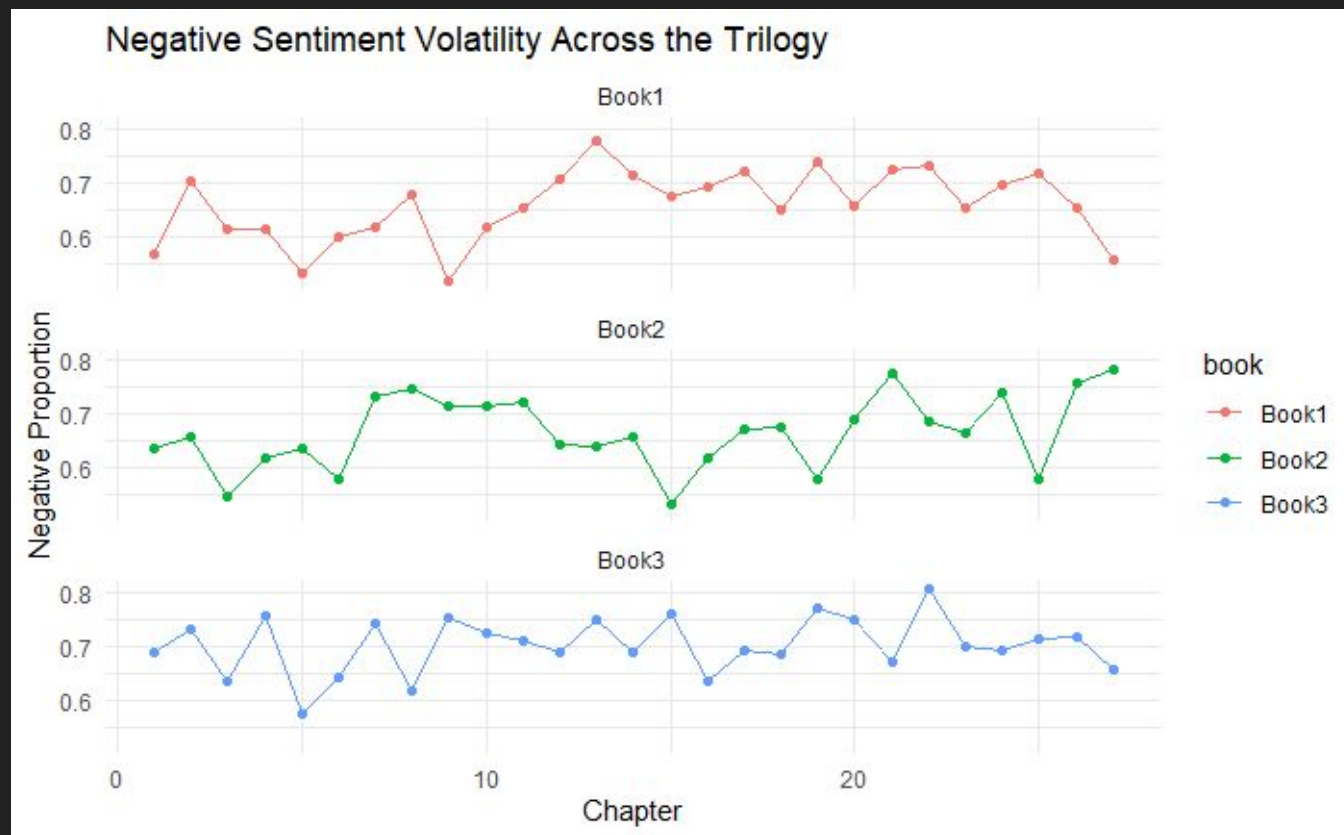
# Writing Style Shift Across the Trilogy



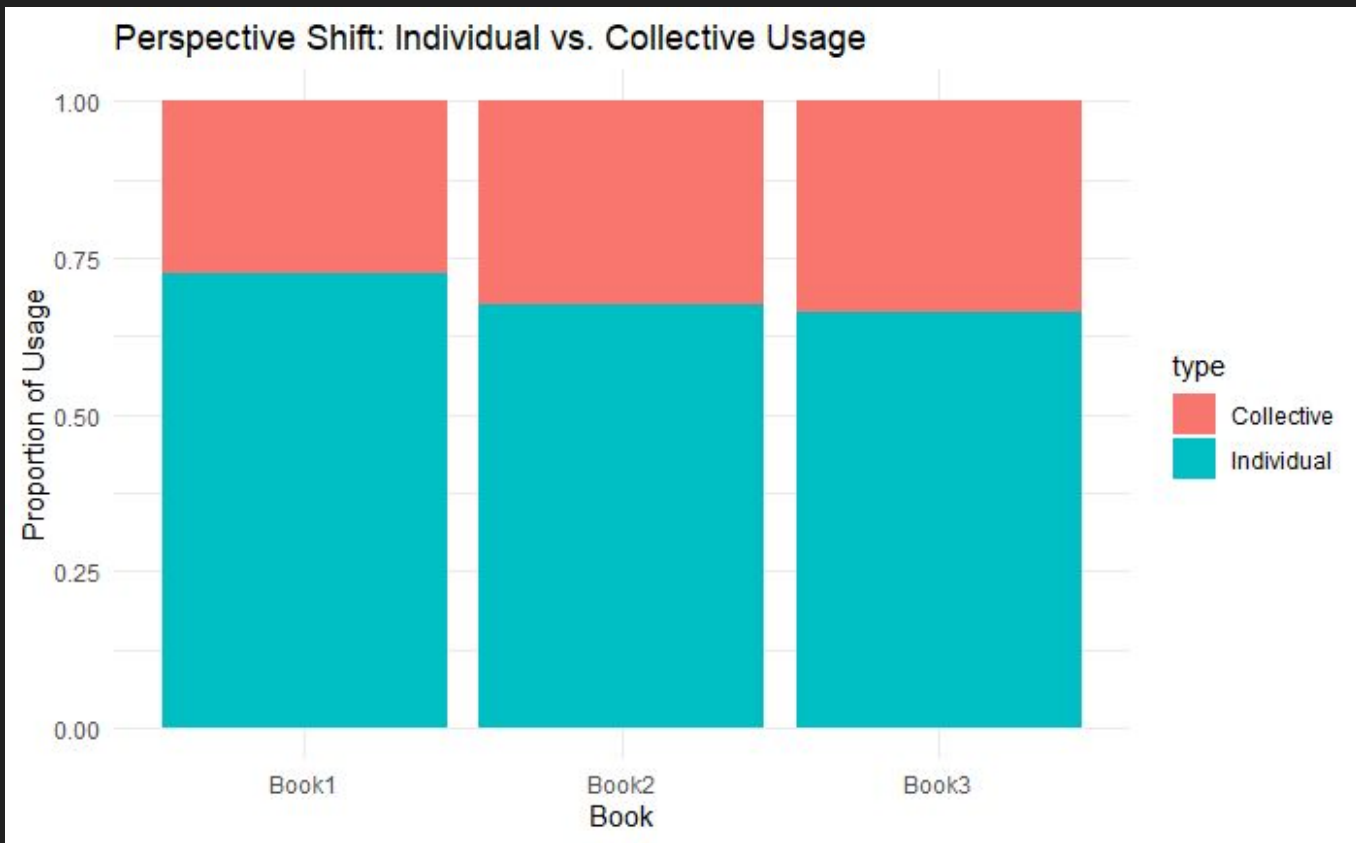
# Sentiment Overview



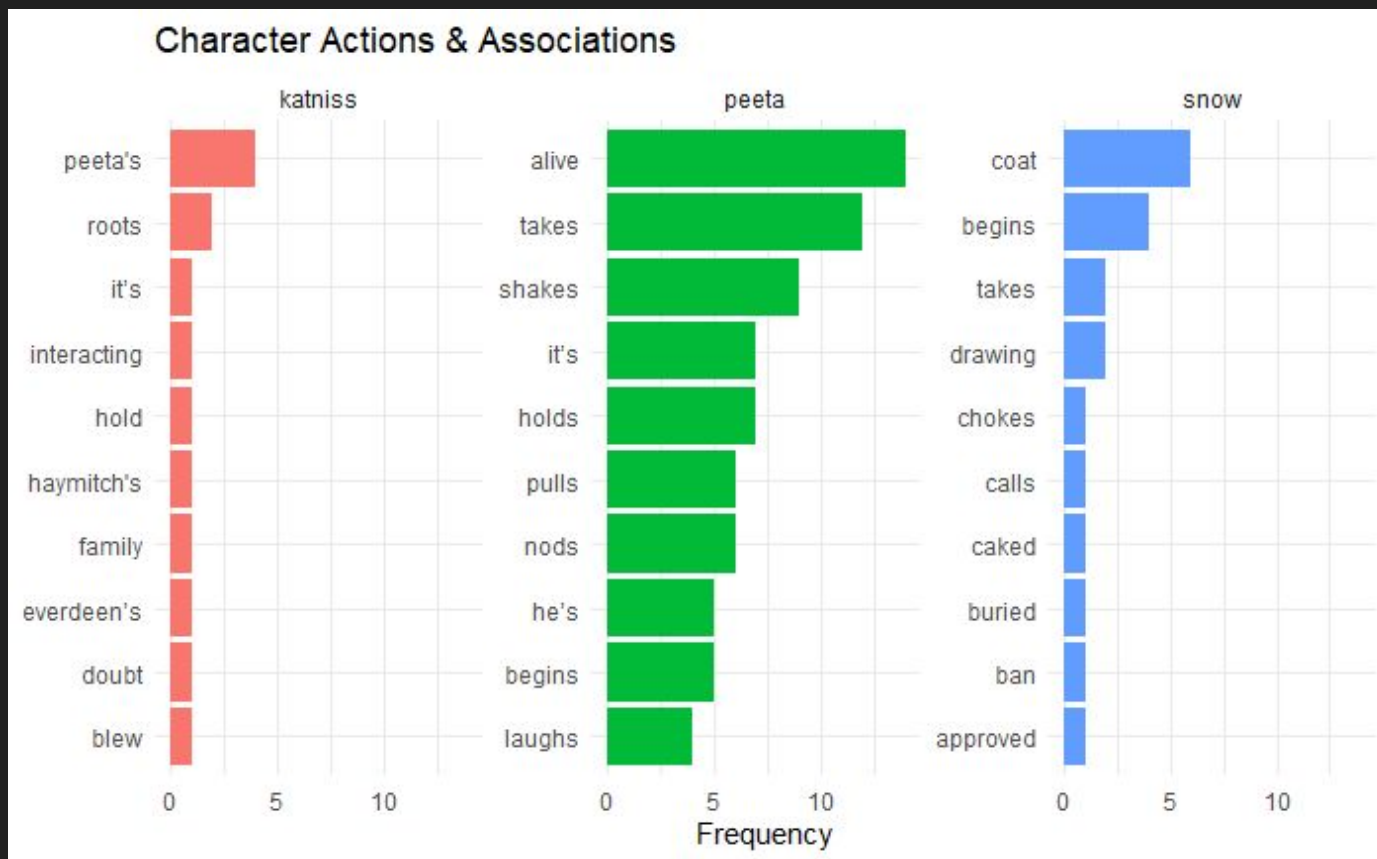
# Sentiment Volatility Comparison (BING)



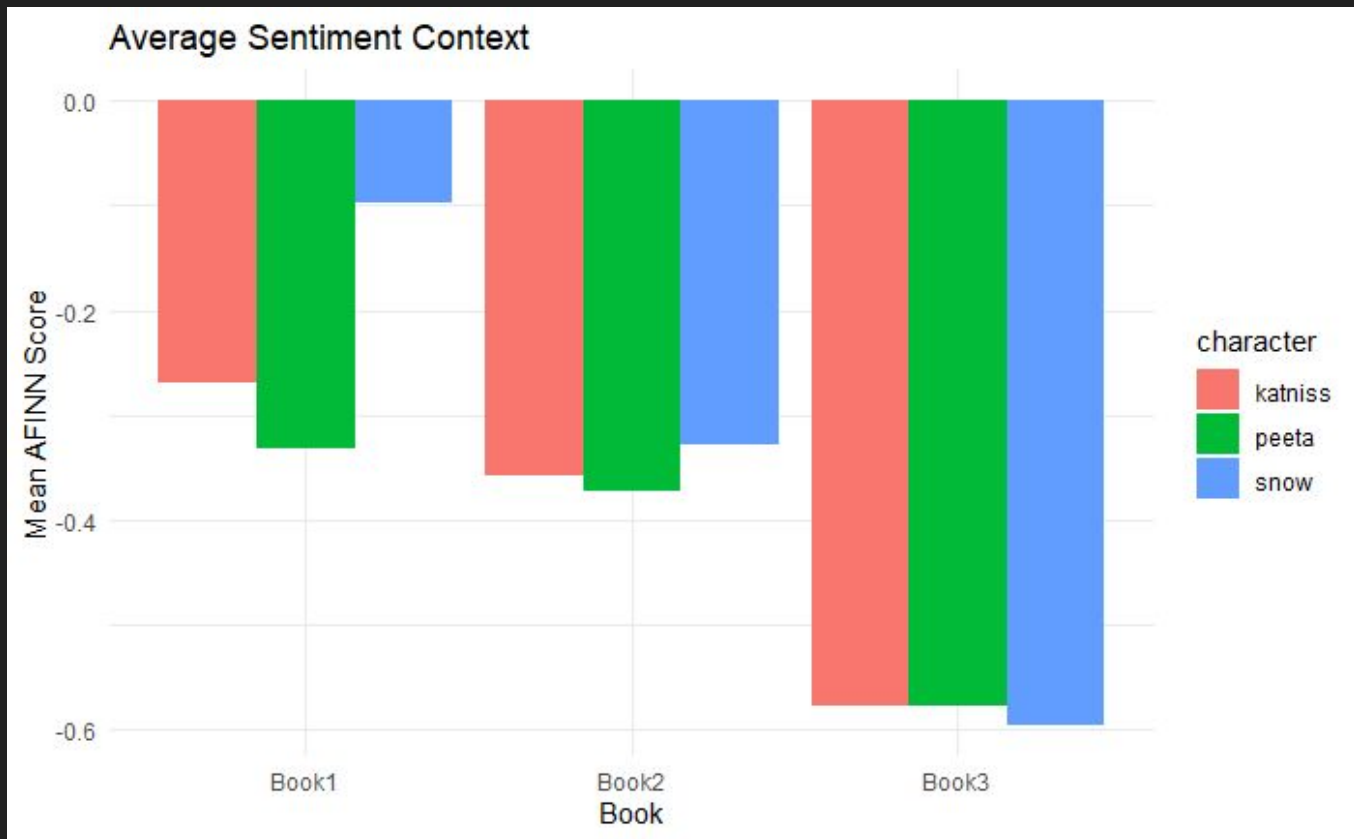
# Pronoun Shift Across the Trilogy



# Bi-Gram Analysis



# Intensity of Sentiment Surrounding Characters (AFINN)

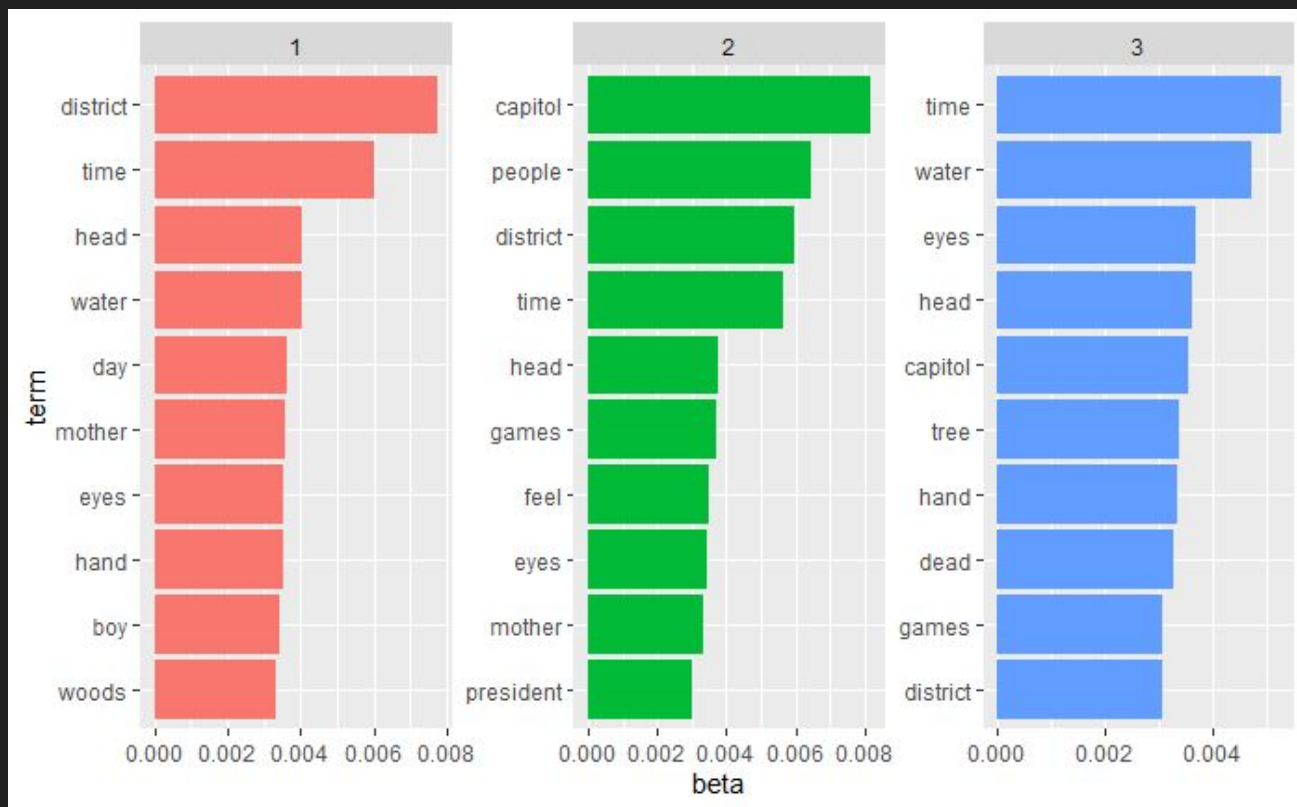


# LDA of Topics in All Books

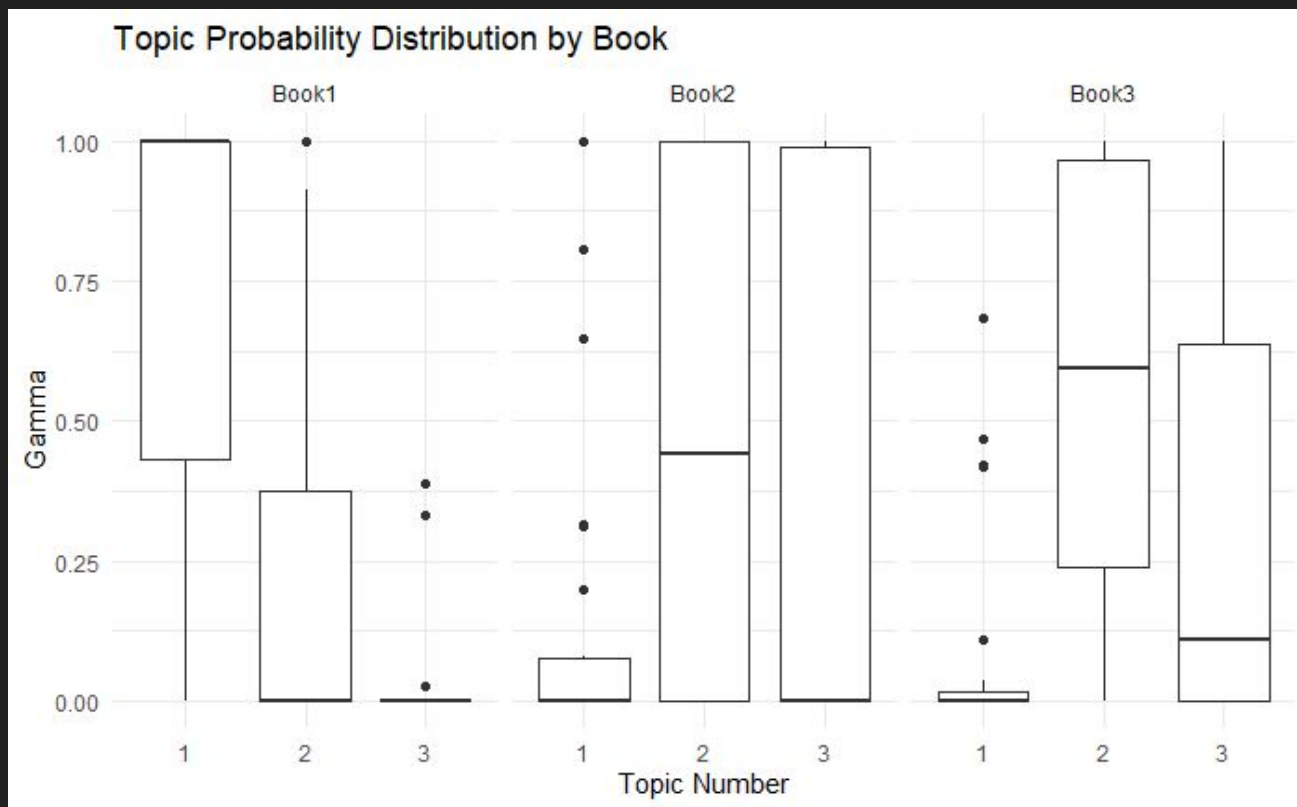
Topic 1:  
Home/District 13  
/Survival

Topic 2:  
Games/Politics  
/People

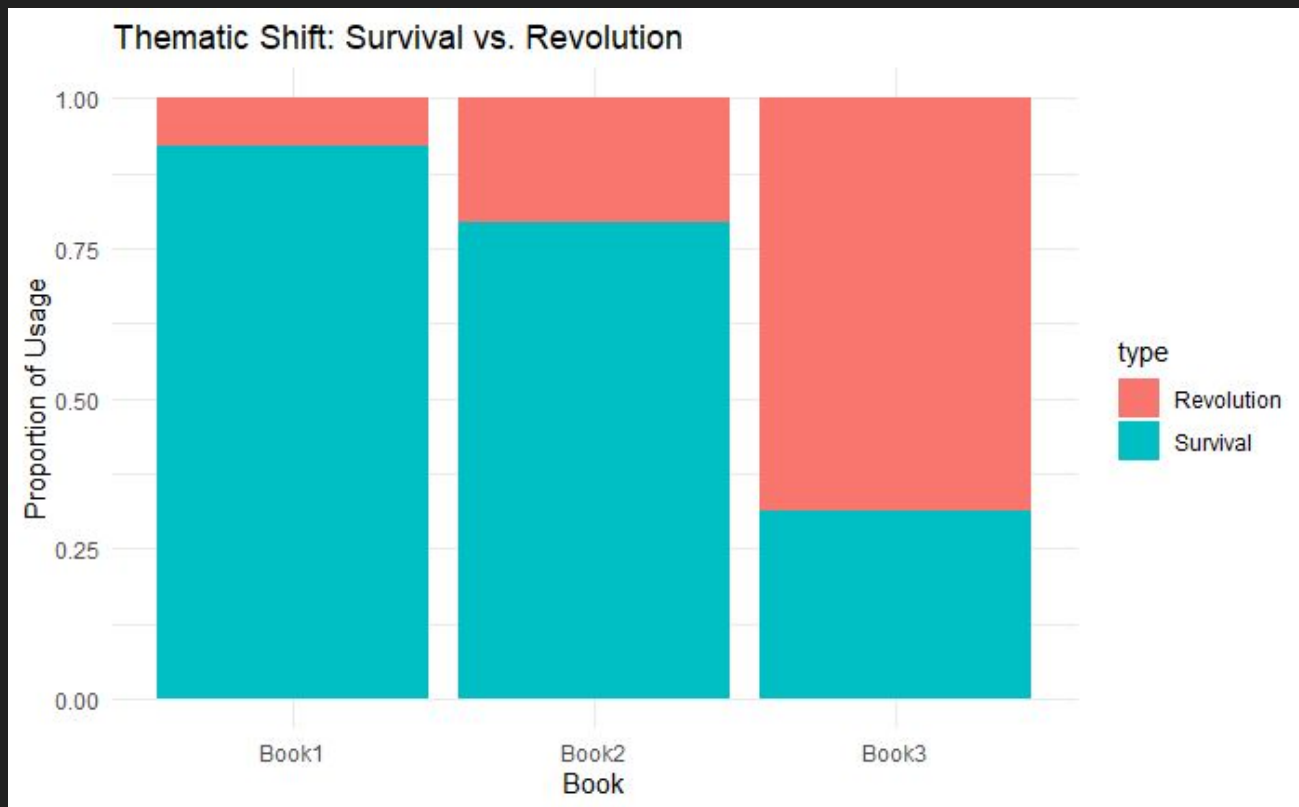
Topic 3:  
Time/Death/Water



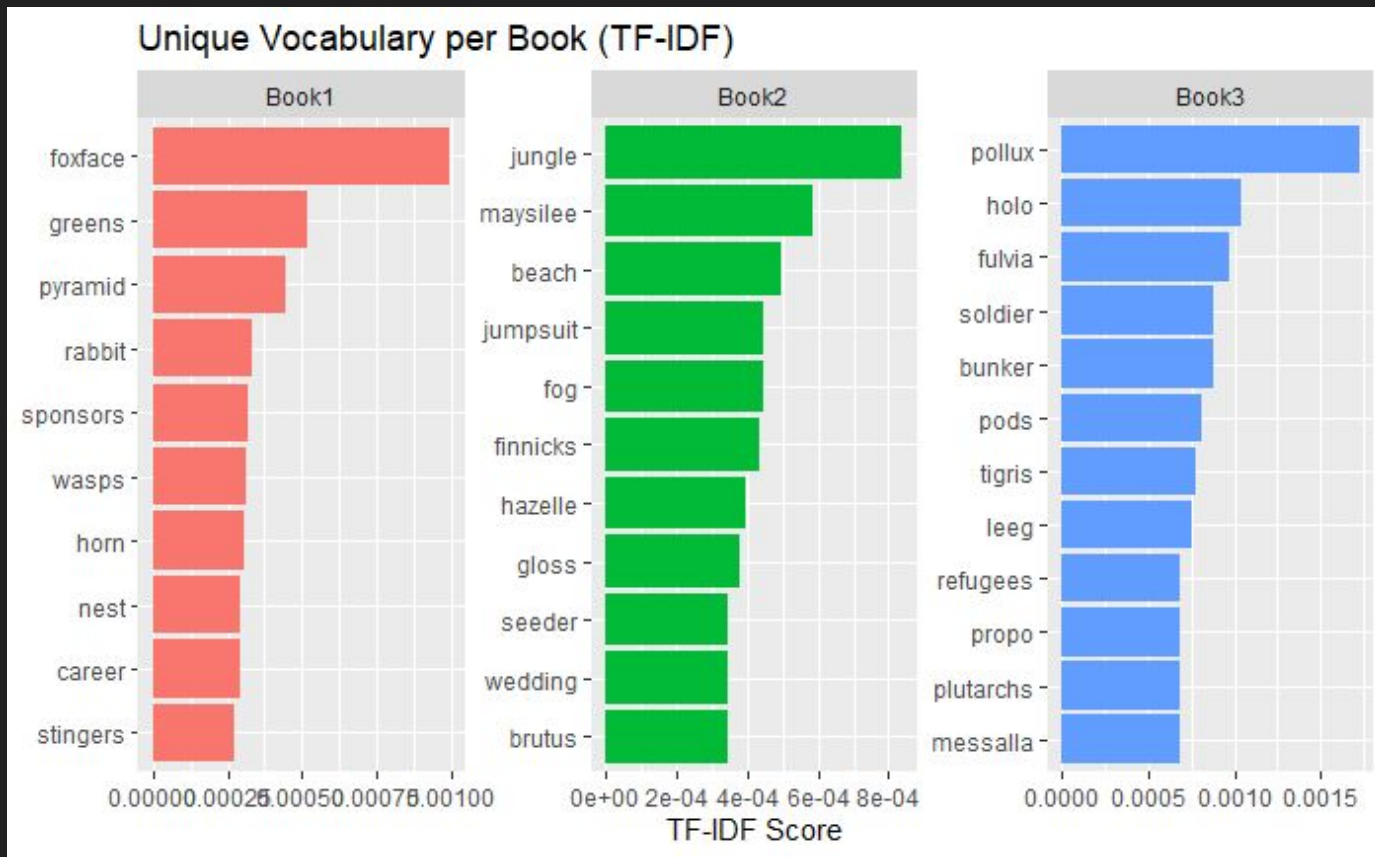
# Topics in Each Book



# Thematic Shift



# Vocabulary Shift Across Trilogy (TF-IDF)





# 06

## Comparison to LLM

# Comparing LLM Unique Words to TF-IDF

- GPT-4 was provided the top 75 most frequently used words per book
  - Tasked with the goal of identifying the key words unique to each book
  - Provided a brief analysis of the words identified (themes)
- Minor overlap with LDA
  - “District” was a key word in book 3, which is a major aspect of topic 1
- Few words captured in our TF-IDF appear here
  - None in book 1
  - “Jungle” in book 2
  - “Pollux” in book 3

07

# Statistical Analysis



# Statistical Testing

We ran various statistical analysis with our data:

- ANOVA: Mean chapter sentiment across books
  - Book ANOVA using BING sentiments (sentiment score  $\sim$  book)
- Chi-Square: Word frequency differences
  - Chi-square of top 50 most used words by book
- Regression: Using Book + Chapter to predict sentiment
  - sentiment score  $\sim$  book + chapter

# Results

## ANOVA:

|           | DF | SUM SQ | MEAN SQ | F VALUE | PR(>F) |
|-----------|----|--------|---------|---------|--------|
| book      | 2  | 4958   | 2479.2  | 3.956   | 0.0231 |
| Residuals | 78 | 48876  | 626.6   |         |        |

## Chi Squared:

| X-SQUARED | DF | P-VALUE  |
|-----------|----|----------|
| 744.48    | 98 | <2.2e-16 |

## Regression:

| COEFFCIENTS    | ESTIMATE | STD. ERROR | T VALUE | PR(> T ) |
|----------------|----------|------------|---------|----------|
| (Intercept)    | -44.1481 | 6.6072     | -6.682  | 3.31e-09 |
| bookBook2      | -0.4444  | 6.4837     | -0.069  | .94553   |
| bookBook3      | -16.8148 | 6.4837     | -2.593  | 0.01137  |
| chapter_number | -1.0265  | 0.3398     | -3.020  | 0.00342  |

| RESIDUAL STD ERROR | MULTIPLE R SQUARED | ADJUSTED R SQUARED | F-STATISTIC | P-VALUE |
|--------------------|--------------------|--------------------|-------------|---------|
| 23.82 (on 77 DF)   | 0.1883             | 0.1567             | 5.953       | 0.00105 |



08

# Interpretations & Conclusions

# Interpretation

- Language relating to fear, trust, and anticipation appear consistently across the trilogy
- Book 3 has the lowest average sentiment score, indicating the darkest tone
- Book 1 shows the most volatile sentiment changes, likely due to the unpredictable emotional states present
- Pronoun usage is largely consistent across books, though Book 3 shows a slight shift to more collective language
- Frequent words and topic modeling show a clear thematic shift across the trilogy
- Word association shows each character is surrounded by distinct language
- Our statistical test results (ANOVA, Chi-square, regression) showed significance

# Limitations

- Data preprocessing steps could have influenced word frequencies and sentiment results.
- Computational text analysis may oversimplify deeper narrative meaning and literary nuance.

# Conclusions

- Our analysis reveals a clear shift in tone, themes, and language across the trilogy
  - The narrative shifts from individual survival in the arena (Book 1) to political unrest (Book 2) and ultimately rebellion and war (Book 3)
- Emotional tone becomes progressively darker, with Book 3 showing the most negative sentiment overall
- Character word associations shows linguistically Katniss symbolizes relationships & loyalty, Peeta symbolizes hope, and Snow symbolizes conflict
- Statistical tests confirm that changes in sentiment, vocabulary, and theme are significant across the books
- TF-IDF was better captured by our analysis than GPT-4 (words were less unique)

# References

Collins, Suzanne. The Hunger Games. Scholastic, 2010.

Collins, Suzanne, and Maria Postema. Catching Fire. Van Goor, 2013.

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Thank You!